

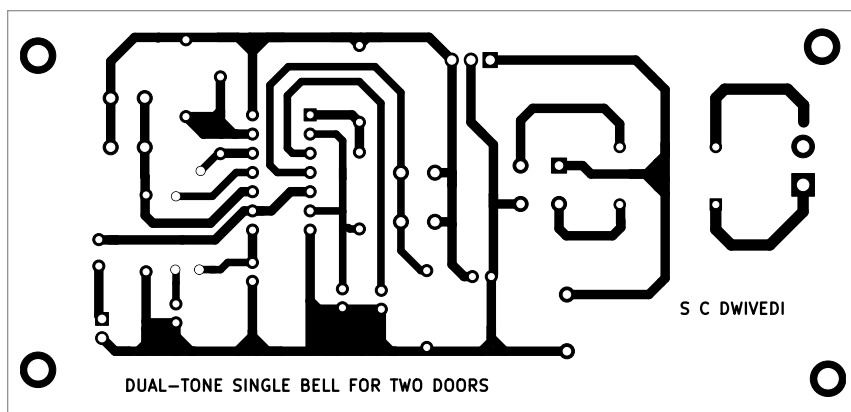


Fig. 3: Actual-size PCB

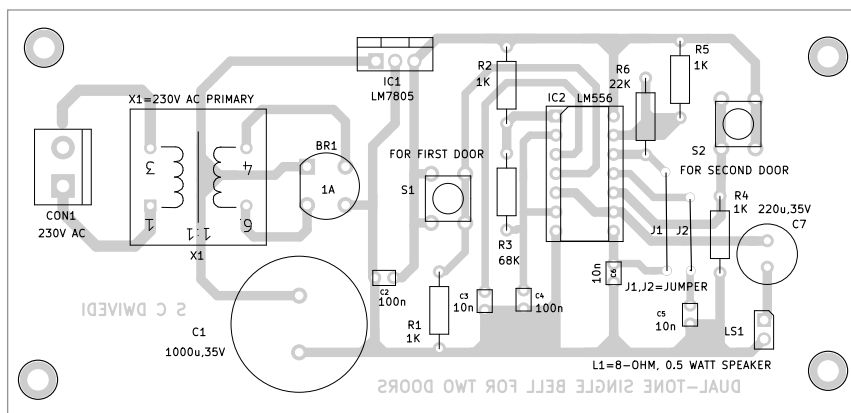


Fig. 4: Component layout

for a deeper tone, while decreasing it raises the frequency for a sharper tone.

The output section, built around a common speaker (LS1) and capacitor C7, produces two distinct tones. The speaker converts the oscillating square-wave signals into audible sounds. Because the timers generate different frequencies, the speaker emits two clearly distinguishable sound patterns. This ensures that, with a single speaker, residents can easily identify which doorbell switch has been pressed, based on whether the sound is higher-pitched or lower-pitched.

Overall, the power supply provides a stable 5V DC to the circuit. Pressing switch S1 triggers IC2A, producing Tone 1, while pressing switch S2 triggers IC2B, generating Tone 2. The speaker then rings at the corresponding frequency. Since both tones are

distinct, the listener can quickly recognise whether the call is from door 1 or door 2. This makes the device simple, practical, and highly efficient for homes, apartments, or offices with more than one entrance.

Construction and testing

An actual-size, single-sided PCB layout for the dual-tone single bell device for two doors is shown in Fig. 3, with its component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box.

For PCB assembly, start with small parts, such as resistors, ceramic capacitors, and jumpers, followed by the rectifier, regulator IC, timer IC, and electrolytic capacitors. Mount connectors for AC input, speaker, and switches, and solder them. House the assembled PCB in a plastic enclosure with external connections for the transformer,

PARTS LIST

Semiconductors:

- IC1 - LM7805, 5V regulator
- IC2 - LM556 timer
- BR1 - 1A bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R4, R5 - 1k Ω
- R3 - 68k Ω
- R6 - 22k Ω

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2, C4 - 100nF ceramic disc
- C3, C5, C6 - 10nF ceramic disc
- C7 - 220 μ F, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- LS1 - 8 Ω , 0.5-watt speaker
- S1, S2 - Bell switch
- X1 - 230V AC primary to 15V AC, 500mA secondary transformer

switches, and speaker.

For testing, first check the regulated 5V output with the LM556 removed. Once verified, insert the LM556 and connect the speaker. Pressing switch S1 should give Tone 1, while S2 produces a distinct Tone 2. If both sound similar, check the resistor and capacitor values. If silent, check the supply and IC connections. After confirming operation, install the switches at the doors and place the speaker centrally.

Installation steps

- Install the door push switches S1 and S2 at the entrances
- Mount the speaker (LS1) at a central location inside the house where it is clearly audible, ensuring it is securely fixed and protected
- Connect the 230V AC input to connector CON1 and keep the cabinet in a suitable place
- Connect the speaker to the output connector LS1
- Insulate all AC parts and test the system. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-dual-tone-single-bell-for-two-doors>

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